

## TIANJIN BINHAI INTL, China

WMO: 545273

Lat: 39.124N

Lon: 117.346E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.9      | -9.0 | -27.2                         | 0.3 | -2.0 | -24.8 | 0.4 | -1.9 | 11.6                  | -2.9 | 10.5 | -3.4 | 3.2                   | 340  | 0.477 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.3                    | 35.0            | 22.8 | 33.8 | 23.1 | 32.1 | 22.8 | 27.5                | 30.7 | 26.7 | 29.9 | 26.0 | 29.1 | 3.7                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.9                            | 22.6 | 29.6 | 26.0 | 21.3 | 28.5 | 25.0 | 20.1 | 27.9 | 87.5          | 31.1 | 83.8 | 30.6 | 80.7 | 28.8 | 31.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 10.2 | 8.7 | 7.4                        | DB  | -13.3              | 37.5 | 2.2                                                | 1.8 | -14.8      | 38.8 | -16.1      | 39.9 | -17.3      | 40.9 | -18.9 | 42.2 |
| (5)               |      |     |                            | WB  | -14.6              | 28.8 | 2.0                                                | 1.0 | -16.1      | 29.5 | -17.2      | 30.1 | -18.4      | 30.7 | -19.8 | 31.4 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual                                                            | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |      |
|------|-------------------------------------------------------------------|-------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|      |                                                                   | (d)                                                               | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg                                                             | 13.5  | -2.9  | 0.5   | 7.4   | 14.9  | 21.1  | 24.9  | 27.4  | 26.4  | 21.8  | 14.6  | 5.9   | -0.8 | (6)  |      |
| (7)  |                                                                   | DBStd                                                             | 11.16 | 3.20  | 3.57  | 4.28  | 3.91  | 3.36  | 2.82  | 2.36  | 2.32  | 3.03  | 3.60  | 4.12  | 2.97 | (7)  |      |
| (8)  |                                                                   | HDD10.0                                                           | 1251  | 401   | 267   | 104   | 6     | 0     | 0     | 0     | 0     | 6     | 133   | 334   |      | (8)  |      |
| (9)  |                                                                   | HDD18.3                                                           | 2725  | 659   | 500   | 339   | 117   | 12    | 1     | 0     | 7     | 124   | 373   | 593   |      | (9)  |      |
| (10) |                                                                   | CDD10.0                                                           | 2527  | 0     | 0     | 23    | 152   | 344   | 448   | 538   | 509   | 355   | 148   | 10    | 0    | (10) |      |
| (11) |                                                                   | CDD18.3                                                           | 959   | 0     | 0     | 1     | 12    | 98    | 198   | 280   | 251   | 112   | 8     | 0     | 0    | (11) |      |
| (12) |                                                                   | CDH23.3                                                           | 9163  | 0     | 0     | 6     | 141   | 884   | 1854  | 2999  | 2431  | 792   | 55    | 0     | 0    | (12) |      |
| (13) |                                                                   | CDH26.7                                                           | 3594  | 0     | 0     | 1     | 28    | 307   | 767   | 1302  | 948   | 234   | 8     | 0     | 0    | (13) |      |
| (14) | Wind                                                              | WSAvg                                                             | 3.2   | 3.1   | 3.2   | 3.8   | 4.2   | 4.0   | 3.4   | 2.9   | 2.6   | 2.7   | 2.9   | 3.0   | 3.0  | (14) |      |
| (15) | Precipitation                                                     | PrecAvg                                                           | 524   | 3     | 6     | 7     | 24    | 37    | 80    | 155   | 121   | 48    | 28    | 13    | 3    | (15) |      |
| (16) |                                                                   | PrecMax                                                           | 776   | 12    | 17    | 44    | 76    | 74    | 154   | 315   | 317   | 124   | 134   | 56    | 11   | (16) |      |
| (17) |                                                                   | PrecMin                                                           | 355   | 0     | 0     | 0     | 4     | 5     | 7     | 71    | 54    | 2     | 1     | 0     | 0    | (17) |      |
| (18) |                                                                   | PrecStd                                                           | 112   | 3     | 6     | 10    | 22    | 20    | 35    | 57    | 54    | 30    | 26    | 15    | 3    | (18) |      |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%                                                              | DB    | 8.8   | 14.0  | 23.2  | 29.9  | 34.8  | 36.2  | 37.2  | 35.1  | 32.9  | 27.2  | 19.8  | 10.1 | (19) |      |
| (20) |                                                                   |                                                                   | MCWB  | 1.5   | 5.8   | 11.5  | 16.0  | 18.4  | 20.0  | 24.0  | 24.5  | 20.8  | 16.1  | 10.6  | 2.9  | (20) |      |
| (21) |                                                                   | 2%                                                                | DB    | 6.0   | 11.2  | 20.1  | 27.0  | 32.0  | 34.2  | 35.1  | 33.8  | 31.0  | 24.9  | 17.0  | 7.8  | (21) |      |
| (22) |                                                                   |                                                                   | MCWB  | 0.4   | 3.8   | 9.9   | 14.9  | 17.8  | 20.5  | 24.2  | 24.4  | 19.8  | 15.2  | 9.3   | 1.9  | (22) |      |
| (23) |                                                                   | 5%                                                                | DB    | 4.1   | 9.1   | 17.2  | 24.9  | 30.0  | 32.8  | 33.9  | 32.2  | 29.2  | 22.8  | 14.8  | 6.0  | (23) |      |
| (24) |                                                                   |                                                                   | MCWB  | -0.8  | 3.0   | 8.5   | 13.8  | 17.3  | 20.9  | 24.5  | 23.9  | 19.6  | 14.6  | 8.2   | 1.2  | (24) |      |
| (25) |                                                                   | 10%                                                               | DB    | 2.8   | 6.9   | 14.9  | 22.8  | 28.0  | 30.9  | 32.2  | 31.1  | 27.9  | 20.9  | 12.2  | 4.2  | (25) |      |
| (26) |                                                                   |                                                                   | MCWB  | -1.5  | 1.7   | 7.2   | 13.0  | 16.6  | 21.0  | 24.4  | 23.7  | 19.2  | 13.6  | 7.1   | 0.2  | (26) |      |
| (27) |                                                                   | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%  | WB    | 2.5   | 7.3   | 13.9  | 18.9  | 22.9  | 26.3  | 28.7  | 28.6  | 25.6  | 20.0  | 13.0 | 5.0  | (27) |
| (28) |                                                                   |                                                                   |       | MCDWB | 6.3   | 11.0  | 20.2  | 26.1  | 28.4  | 30.5  | 32.4  | 31.3  | 28.8  | 23.1  | 15.6 | 6.6  | (28) |
| (29) |                                                                   |                                                                   | 2%    | WB    | 1.0   | 5.1   | 11.5  | 17.2  | 21.2  | 24.7  | 27.8  | 27.5  | 23.8  | 18.5  | 10.8 | 3.1  | (29) |
| (30) |                                                                   |                                                                   |       | MCDWB | 4.4   | 8.7   | 17.5  | 23.3  | 27.0  | 28.9  | 31.1  | 30.3  | 26.5  | 21.4  | 14.3 | 5.6  | (30) |
| (31) | 5%                                                                |                                                                   | WB    | 0.0   | 3.8   | 9.8   | 15.8  | 20.0  | 23.8  | 27.0  | 26.6  | 22.6  | 17.0  | 9.5   | 2.0  | (31) |      |
| (32) |                                                                   |                                                                   | MCDWB | 2.9   | 7.8   | 15.4  | 21.8  | 25.9  | 28.1  | 30.4  | 29.5  | 25.5  | 19.6  | 13.0  | 4.4  | (32) |      |
| (33) | 10%                                                               |                                                                   | WB    | -1.0  | 2.4   | 8.2   | 14.4  | 18.8  | 22.9  | 26.3  | 25.9  | 21.4  | 15.5  | 8.2   | 1.0  | (33) |      |
| (34) |                                                                   |                                                                   | MCDWB | 2.0   | 6.4   | 13.2  | 20.0  | 24.6  | 27.1  | 29.6  | 28.7  | 24.8  | 19.0  | 11.2  | 3.5  | (34) |      |
| (35) | Mean Daily Temperature Range                                      |                                                                   | 5% DB | MDBR  | 9.3   | 10.3  | 10.9  | 11.2  | 11.0  | 9.8   | 8.3   | 8.4   | 9.8   | 10.2  | 9.5  | 8.6  | (35) |
| (36) |                                                                   |                                                                   |       | MCDBR | 11.3  | 13.2  | 14.3  | 14.6  | 13.9  | 12.5  | 10.5  | 9.7   | 11.6  | 12.2  | 12.4 | 10.9 | (36) |
| (37) |                                                                   |                                                                   |       | MCWBR | 7.0   | 7.5   | 6.8   | 6.0   | 5.2   | 4.5   | 4.0   | 3.9   | 4.3   | 5.3   | 6.5  | 6.7  | (37) |
| (38) |                                                                   |                                                                   | 5% WB | MCDWB | 10.4  | 12.2  | 13.2  | 12.9  | 11.4  | 9.8   | 8.3   | 8.1   | 9.5   | 10.0  | 10.9 | 9.2  | (38) |
| (39) |                                                                   | MCWBR                                                             |       | 6.9   | 7.6   | 7.0   | 6.6   | 5.6   | 4.6   | 4.2   | 4.1   | 4.8   | 5.6   | 6.7   | 6.6  | (39) |      |
|      |                                                                   |                                                                   |       |       |       |       |       |       |       |       |       |       |       |       |      |      |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub                                                              | 0.468 | 0.544 | 0.564 | 0.631 | 0.675 | 0.806 | 0.806 | 0.755 | 0.680 | 0.653 | 0.535 | 0.484 | (40) |      |      |
| (41) |                                                                   | taud                                                              | 1.869 | 1.741 | 1.725 | 1.645 | 1.595 | 1.430 | 1.471 | 1.542 | 1.615 | 1.620 | 1.799 | 1.884 | (41) |      |      |
| (42) |                                                                   | Ebn at Noon                                                       | 641   | 644   | 690   | 677   | 660   | 577   | 572   | 585   | 593   | 543   | 567   | 586   | (42) |      |      |
| (43) |                                                                   | Edh at Noon                                                       | 146   | 191   | 214   | 246   | 263   | 311   | 296   | 269   | 236   | 213   | 157   | 136   | (43) |      |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg                                                            | 2.37  | 3.08  | 4.47  | 5.32  | 5.95  | 5.31  | 4.79  | 4.66  | 4.17  | 3.19  | 2.41  | 2.10  | (44) |      |      |
| (45) |                                                                   | RadStd                                                            | 0.24  | 0.36  | 0.40  | 0.28  | 0.40  | 0.51  | 0.35  | 0.29  | 0.34  | 0.32  | 0.34  | 0.23  | (45) |      |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.56 | N/A     | N/A    | N/A     | -0.42 | -0.78 | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (9 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +80     | +61     |

Nomenclature: See separate page